PHYS 225

Week 12, Lecture 10:
Dynamics with 4-vectors and the Doppler effect

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Reminders

- **Supplementary reading** is posted on the website/**schedule** HW/**reading**
  Rapidity (Week #**7**).
  Relativistic Force in 2D (Week #**10**).
  Nabla in general coordinates (Week #**11**).

- Finals are scheduled! Please search for possible conflicts early.

  Rule to ask for conflict final exam:

  https://studentcode.illinois.edu/article3/part2/3-201/
  If you believe you have a conflict with your PHYS 225 final exam and the other (A1 or A2) exam won't resolve your conflict, please send an e-mail to: Mrs. Kate Shunk (shunk@Illinois.edu)
  In your e-mail provide: Your name, Your netID, PHYS 225, A copy of your final exam schedule: https://courses.illinois.edu/user/student/courselist/2026/spring

- Office Hours Wednesday, Thursday usual time.
- Homework 10 assigned this Thurs. 4/9 due next Thurs. 4/16.
- Don't forget to turn in your **1-minute papers** by 6 pm today!!!

Week 12, Lecture 10

Dynamics with 4-vectors; Doppler effect

Objectives:

- Use natural units to simplify relativistic calculations
- Construct the displacement, velocity, energy-momentum, acceleration, and force 4-vectors and understand their transformation properties
- Characterize energy-momentum 4-vectors as massive or massless using the relativistic dot product
- Derive the longitudinal Doppler effect using Lorentz transformations

One Minute Paper: remind yourself of the Doppler effect in classical mechanics. If a fire truck is driving towards you, do you hear the siren at a higher or lower frequency?

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Natural units

From now on in the course, we will choose units such that c = 1.

$$v = 1/2 \implies v = c/2$$

$$E_0 = m \implies E = mc^2$$

If we measure time in seconds, this is equivalent to measuring distances in light-seconds. We actually do this all the time! “I’m 15 minutes away.”

Can always restore c by dimensional analysis (Morin 6.6.2)

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Newtonian vectors

What is a vector in Newtonian mechanics?

(Discussion 3, Discussion 9): an ordered triple of numbers v^i that transforms the same way as the coordinates under rotations

$$x'^i = R^i{}_j x^j \iff v'^i = R^i{}_j v^j$$

We can build vectors from other vectors, as long as we perform operations that are **invariant** under rotations:

$$\mathbf{x}(t) \rightarrow \mathbf{v} = \frac{d\mathbf{x}(t)}{dt} \rightarrow \mathbf{a} = \frac{d^2\mathbf{x}(t)}{dt^2} \rightarrow \mathbf{F} = m\mathbf{a}$$

position | velocity | acceleration | force

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Relativistic 4-vectors

What is a 4-vector in special relativity?

An ordered 4-tuple of numbers v^μ that transforms the same way as the coordinates under Lorentz transformations:

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu \Longleftrightarrow v'^\mu = \Lambda^\mu{}_\nu v^\nu$$

We can follow the same steps to construct vectors from the displacement 4-vectors, but time is no longer invariant! So we have to differentiate with respect to **proper time.**

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Displacement 4-vector and proper time

(Infinitesimal) displacement 4-vector:

$$dx^\mu = (dt, dx, dy, dz)$$

(See Lecture 7)

(Infinitesimal) proper time is Lorentz-invariant:

$$(d\tau)^2 \equiv dx_\mu dx^\mu = (dt)^2 - (dx)^2 - (dy)^2 - (dz)^2$$

Remember, this is the time elapsed in a frame which travels the distance $d\mathbf{x} = (dx, dy, dz)$ in a time dt

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$$\boxed{g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}}$$

$dx^0 dx^0 = 1$

$dx^1 dx^1 = -1$

$-\ -\ -\ -$

$$\underline{\Delta S^2} = g_{\mu\nu}\,\Delta x^\mu \Delta x^\nu > 0 \qquad c = 1$$

$$ds^2 = d\tau^2 = dt^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

$x \quad y \quad z$

$$\underline{J_\mu J^\mu} = (J^0 J^0) - (\vec{J})^2$$

$\rho \quad \varphi \quad \theta$

$-\ -\ -\ -\ -\ -$

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$$\frac{dx^\mu}{d\tau} = \frac{dx^\mu}{dt} \cdot \frac{dt}{d\tau}$$

Velocity and energy-momentum 4-vectors

4-vector

$$V^\mu \equiv \frac{dx^\mu}{d\tau} = \left(\frac{dt}{d\tau}, \frac{d\mathbf{x}}{dt}\frac{dt}{d\tau}\right)$$

invariant

ordinary velocity **v**

convenient notation: write 3 spatial components as an (ordinary) vector

$$d\tau = dt\sqrt{1-(d\mathbf{x}/dt)^2} = dt/\gamma$$

(time dilation)

$$\Longrightarrow \boxed{V^\mu = (\gamma, \gamma\mathbf{v}) \quad \text{velocity 4-vector}}$$

Mass is invariant: $\boxed{P^\mu \equiv mV^\mu = (\gamma m, \gamma m\mathbf{v}) \quad \text{energy-momentum 4-vector}}$

$$d\tau^2 = dt^2 - d\vec{x}^2 \longrightarrow 1 = \left(\frac{dt}{d\tau}\right)^2 - \left(\frac{d\vec{x}}{d\tau}\right)^2$$

$$d\tau^2 = dt^2\left(1 - \frac{d\vec{x}^2}{dt^2}\right)$$

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Acceleration 4-vector

Let's go back to the velocity and differentiate one more time.

$$V^\mu = (\gamma, \gamma \mathbf{v})$$

remember: this is the gamma factor between the particle's rest frame and the frame with velocity **v**

$$A^\mu \equiv \frac{dV^\mu}{d\tau} = \gamma \frac{dV^\mu}{dt} = \gamma \left(\frac{d\gamma}{dt}, \frac{d(\gamma \mathbf{v})}{dt} \right)$$

Recall from lecture 8: $\dfrac{d\gamma}{dt} = \gamma^3 v \dot{v}$

$$A^\mu = (\gamma^4 v \dot{v}, \gamma^4 v \dot{v} \underline{\vec{\mathbf{v}}} + \gamma^2 \vec{\mathbf{a}})$$

$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$

V

$\vec{V} = V_x \hat{x} + V_y \hat{y} + V_z \hat{z}$

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Force 4-vector

Like we did in Lecture 8, let's define 4-force as the derivative of 4-momentum, but with respect to **proper** time:

$$F^\mu \equiv \frac{dP^\mu}{d\tau}$$

Newton's 2nd Law now appears as 4-vector equation:

$$F^\mu = \frac{d(mV^\mu)}{d\tau} = m\frac{dV^\mu}{d\tau} = mA^\mu$$

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$c = 1$

Energy-momentum conservation

Our concepts of relativistic energy and momentum from Lecture 8 are encoded in the energy-momentum 4-vector:

$$P^\mu = (\gamma m, \gamma m \mathbf{v})$$

$$E \equiv P^0 = \gamma m$$

$$\mathbf{p} \equiv P^i = \gamma m v^i$$

when dealing with 4-vectors, Latin indices (i, j, k, etc.) used to label spatial components

Energy conservation
Momentum conservation

$\rightarrow$ Total energy-momentum 4-vector is conserved

(**much** more on this next week!)

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Energy-momentum relation for massive particles

A massive particle can always be brought to rest. In its rest frame, its energy is its rest energy and its momentum is zero:

$$P^\mu = (E, \mathbf{p}) \xrightarrow{\text{boost to rest frame}} P'^\mu = (m, 0, 0, 0)$$

Now use invariance of the relativistic dot product:

$$P \cdot P = P' \cdot P'$$

$$E^2 - \mathbf{p}^2 = m^2$$

Same as derived in Discussion 8!

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Energy-momentum transformation laws

Now that we know P is a 4-vector, we can immediately write down how it transforms between frames.

Let's assume momentum only has an x-component, and S' is moving along the x-axis:

$$\begin{pmatrix} E \\ p_x \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} E' \\ p'_x \end{pmatrix}$$

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Energy-momentum transformation laws

A very common situation is when S' is the rest frame of a particle of mass m.

$$\begin{pmatrix} E' \\ p'_x \end{pmatrix} = \begin{pmatrix} m \\ 0 \end{pmatrix}$$

This helps to remember the signs: if we boost this particle in the x-direction, it acquires positive x-momentum

$$\begin{pmatrix} E \\ p_x \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} m \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma m \\ \gamma\beta m \end{pmatrix} \quad = \gamma m v$$

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Massive and massless 4-vectors

Energy-momentum 4-vectors for massive particles are timelike: $P_\mu P^\mu = m^2 > 0$

Spacelike P would imply negative energy: unphysical

What about lightlike?

$P_\mu P^\mu = 0 \longrightarrow P^\mu$ describes a **massless** particle

We can't derive this from V like we did for massive particles since proper time vanishes for a lightlike 4-vector...

$$P_\mu P^\mu = (p^0)^2 - (\vec{p})^2$$

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Massive and massless 4-vectors

…so we just follow our noses.

$$P^\mu = (E, \mathbf{p}) \rightarrow P_\mu P^\mu = E^2 - \vec{p}^{\,2}$$

$$P_\mu P^\mu = 0 \quad \longrightarrow \quad E = |\mathbf{p}|$$

To use this to describe photons, we need to import one fact from quantum mechanics: $E = hf = hc/\lambda$

$$\Longrightarrow \quad P^0 \propto f \propto 1/\lambda$$

Boosting a massless 4-vector doesn't change its velocity, but instead changes its **frequency** (and wavelength)

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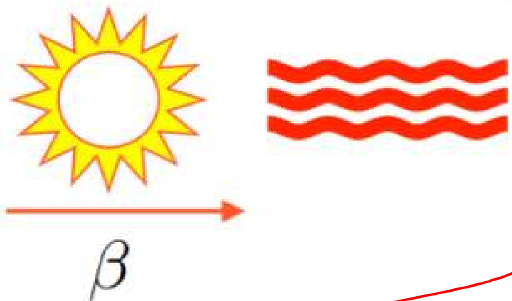
$$E_{obs} = E_{emit}\,\gamma(1+\beta)$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{(1+\beta)(1-\beta)}}$$

Longitudinal Doppler effect



Photons emitted with frequency f' in moving object's rest frame, along direction of motion. What frequency does an observer measure?

[blackboard]

$$\frac{f_{\text{obs}}}{f_{\text{emit}}} = \sqrt{\frac{1+\beta}{1-\beta}}$$

$\beta > 0$: photon frequency increases, wavelength decreases: **blueshift**

$\beta < 0$: photon frequency decreases, wavelength increases: **redshift**

$$E_{obs} = E_{emit}\sqrt{\frac{1+\beta}{1-\beta}}\ ; \quad E \sim f$$

Longitudinal Doppler effect

Say, photons emmited in x direction with energy E_{emit}

$$p'^{\mu} = (E_{emit}, E_{emit})$$

remember $P_x = E_{emit}$

(S' object's rest frame)

Lorentz transformations along x:

$$p^{\mu} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} E_{emit} \\ E_{emit} \end{pmatrix} = \begin{pmatrix} \gamma E_{emit}(1+\beta) \\ \gamma E_{emit}(1+\beta) \end{pmatrix}$$

(what we observe)

New Energy then:

$$E_{obs} = E_{emit}\gamma(1+\beta) = E_{emit}\frac{1+\beta}{\sqrt{1-\beta^2}} =$$

$$= E_{emit}\sqrt{\frac{(1+\beta)^2}{(1-\beta)(1+\beta)}} = E_{emit}\sqrt{\frac{(1+\beta)}{(1-\beta)}}$$

So

$$\frac{f_{obs}}{f_{emit}} = \frac{E_{obs}}{E_{emit}} = \sqrt{\frac{1+\beta}{1-\beta}}.$$

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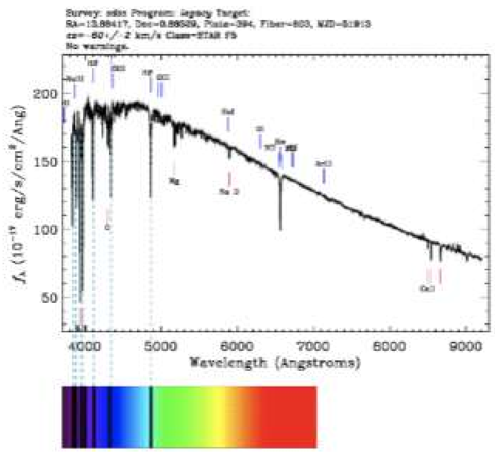
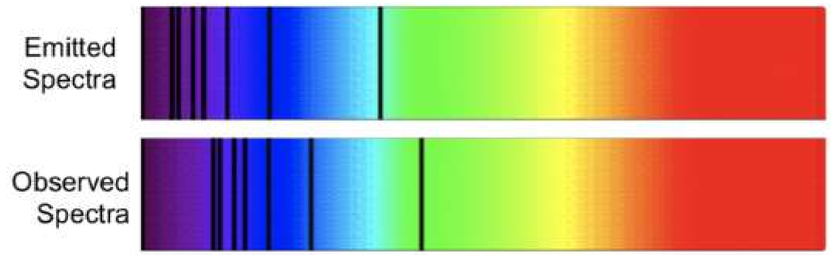
Application: Doppler effect in astrophysics

Quantum mechanics teaches us that atoms and molecules emit light at specific, **quantized**, wavelengths.
If we expect a certain wavelength but see a slightly different one, we can determine the velocity of an object!





much more practice in discussion and on HW!

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Dynamics with 4-vectors; Doppler effect

Summary

- Relativity is easier without c's
- Newton's Second Law generalizes to a 4-vector equation
- Massive particles are described by a timelike energy-momentum 4-vector, which transforms between frames with a Lorentz matrix
- Massless particles (photons) are described by lightlike energy-momentum 4-vectors; their velocity is always the same but their energy (and frequency, wavelength) changes between frames

One Minute Paper: if a star emits light at a wavelength of 550 nm, but you detect the light at 540 nm, is the star moving toward you or away from you?